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Integrated M.C.A. (First Semester)

Examination, Dec. 2022

IMCA-104

MATHEMATICS-I

Time Allowed : Three Hours

Maximum Marks : 70

Min. Pass Marks : 28

*This question paper contains **three** sections as under :*

Section-A

Max. Marks-10

*This section shall have **one** compulsory question comprising **10** questions (maximum **20** words answer) taking **two** questions from each unit. Each question shall be of **one** mark and total marks of this section will be **10**. This section will be compulsory in the paper.*

Section-B**Max. Marks-25**

This section will carry 25 marks with equally divided into five long answer type questions (answer in about 250 words) and students are instructed to attempt five questions by selecting one question from each unit.

Section-C**Max. Marks-35**

Will carry 35 marks with five long answer type questions comprising one compulsory question (Question No. 7) of 15 marks and four questions of 10 marks each. Students are instructed to attempt total three questions with one compulsory question (answer about in 500 words) and any two more questions (answer about in 400 words) out of remaining four questions.

Section A

1. (i) State the Euler's theorem.
- (ii) Define the envelope.

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[Contd....

- (iii) Define the saddle point.
- (iv) Define the stationary point.
- (v) State the fundamental theorem of integral calculus.
- (vi) Define the rectification
- (vii) Define the rank of a matrix.
- (viii) Define the characteristic equation of a matrix.
- (ix) State the Cayley-Hamilton theorem.
- (x) Define the eigen value and eigen vector.

Section B**UNIT-I**

2. At any point P of the parabola $y^2 = 4ax$ prove that :

$$\rho = \frac{2}{\sqrt{a}} (x+a)^{3/2} = \frac{2(SP)^{3/2}}{\sqrt{a}}$$

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$[1 + (y'_{cn})^2]^{3/2}$ [Contd....

3. Show that the asymptotes of the following cubic cut the curve again in three points which lie on the straight line $x - y + 1 = 0$,

$$x^3 - 2y^3 + xy(2x - y) + y(x - y) + 1 = 0.$$

UNIT-II

4. Find the points where the function $x^3 + y^3 - 3axy$ has max. or min. value.
5. If

$$u = f(y - z, z - x, x - y)$$

then prove that :

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0.$$

UNIT-III

6. Evaluate :

$$\int_0^a x^4 \sqrt{a^2 - x^2} dx.$$

7. Find the whole area of the Astroid :

$$x = a \cos^3 \theta, \quad y = b \sin^3 \theta.$$

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UNIT-IV

8. Show that the following vectors are L.I. :

$$\alpha_1 = (1, 1, 0); \quad \alpha_2 = (1, 1, 1); \quad \alpha_3 = (2, 1, 3).$$

9. Find the rank of matrix A where :

$$A = \begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 7 \end{bmatrix}.$$

UNIT-V

10. Find the eigen values and corresponding eigen vectors of the matrix A :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{bmatrix}.$$

11. Find the characteristics equation of the following matrix A and verify that it is satisfied by A :

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{bmatrix}.$$

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Section C

12. For what value of λ and μ , the following system of equations :

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

have :

- (i) a unique solution
- (ii) an infinite number of solutions
- (iii) no solution.

13. Trace the following cissoid :

$$y^2(2a - x) = x^3.$$

14. If $x = r \cos \theta$, $y = r \sin \theta$, then prove that :

$$(i) \left(\frac{\partial r}{\partial x} \right)^2 + \left(\frac{\partial r}{\partial y} \right)^2 = 1$$

$$(ii) \frac{\partial r}{\partial x} = \frac{\partial x}{\partial r}$$

$$(iii) \frac{\partial x}{\partial \theta} = r^2 \frac{\partial \theta}{\partial x}$$

$$(iv) \frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} = \frac{1}{r}$$

15. Show that the following matrix A satisfies Cayley Hamilton theorem. Hence find its inverse A^{-1} :

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

16. Find the volume and surface area of the solid generated by the cardioid $r = a(1 + \cos \theta)$ when it revolves about the initial line.